Model Selection and Feature Analysis

PlaySafe

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1.Introduction

PlaySafe is a machine learning solution designed to estimate whether a professional football player will experience at least one recorded injury during the following LaLiga season. The purpose of the solution is to support data-driven decision-making by providing information that sports scientists, physical trainers, medical staff and technical analysts can use when assessing player availability and planning training or squad management.

Football injuries can affect player availability and team performance. By analysing historical information about players, their playing exposure and their injury records, PlaySafe aims to identify patterns associated with injuries in subsequent seasons.

2.AI Solution Architecture

2.1 Architecture overview

PlaySafe follows a supervised machine learning approach. Historical player-season records are processed to create a structured dataset, which is then used to train and evaluate classification models. The selected model can subsequently generate predictions for player-season records containing the required input features.

Each stage has a specific purpose within the overall architecture.

1. Historical data

The solution uses historical LaLiga player and injury data collected from a non-official source. The available information includes player characteristics, playing exposure and recorded injury history.

2. Data preprocessing

The raw data requires cleaning before it can be used for machine learning. This includes standardising formats, converting variables to appropriate data types, handling missing values and preparing and scaling categorical and numerical features.

3. Feature engineering

The original dataset contains injury-level records. These records are aggregated by player and season to create a dataset with one row per player-season. Season-level variables are calculated or prepared, and the target variable is created to indicate whether the player has at least one recorded injury in the following season.

4. Model training

The prepared dataset is used to train Logistic Regression and Random Forest classifiers. A k-Nearest Neighbours (k-NN) model was also used during the initial modelling iteration.

5. Model evaluation

The models are evaluated using a chronological train-test split. Earlier seasons are used for training, while later seasons are reserved for testing. This approach assesses performance on seasons that occur after the training period.

6. Risk prediction

The prototype uses the trained models to estimate injury risk for a player-season record. Random Forest is used to assign the low, medium or high risk category in the current MVP. These categories are based on predefined probability thresholds and are intended to make the output easier to interpret.

3.Feature Selection and Engineering

3.1 Feature Selection

The selected features were chosen based on their relevance to player characteristics, playing exposure and recorded injury history. These variables provide different types of information that may help the models distinguish between player-season records with different subsequent injury outcomes.

Features selected for the model are: height, position, age_season, minutes_played, n_injuries, days_out and matches_missed.

Previous injury history was found to have supporting evidence as an injury risk factor (McCall et al., 2015), and reinjuries were found to cause longer absences than other injuries (Ekstrand et al., 2011).
Playing exposure is relevant when studying injury occurrence (Ekstrand et al., 2011). The relationship between workload and injuries is complex (Jiang et al., 2022), and minutes played is not a detailed workload measure, but it gives an approximation of exposure.

The player's name feature is seen as an identifier but is not used as a predictive feature. Using a name as an input could allow the model to learn player-specific associations rather than generalisable patterns.

The selected features combine physical characteristics, playing exposure and injury-related information. However, the available dataset does not include every factor that may influence injuries, such as detailed training loads, medical assessments or information about training intensity. Consequently, the selected features represent the information available for this prototype rather than a complete description of injury risk.

3.2 Feature Engineering

The raw dataset contains individual injury records, whereas the prediction task requires a player-season-level dataset. Therefore, multiple records belonging to the same player and season were grouped to create one observation per player-season.

This transformation makes it possible to represent a player's characteristics, playing exposure and injury history within a consistent seasonal structure.

The target variable, has_injury_next_season, was created as a binary classification target:

- 1: The player has at least one recorded injury in the following season.
- 0: The player has no recorded injury in the following season.

Player-season observations without an available following season outcome were excluded because their target value could not be established from the available data.

It is important to remark that the system predicts whether an injury is recorded in the next season; it does not predict the exact date, severity or number of future injuries.

4.Model Selection and Design

4.1 Initial Model: k-Nearest Neighbours

The first modelling iteration used k-Nearest Neighbours (k-NN) as an initial classification model. This provided a starting point for evaluating whether the available features could support the prediction task.

The k-NN model therefore served as an initial reference point for further model development.

4.2 Final Iteration algorithms

Logistic Regression and Random Forest were selected because they offer different modelling approaches, allowing the project to investigate whether a relatively simple classification model or a more flexible ensemble model would be more suitable for PlaySafe.

4.3 Logistic Regression

Logistic Regression was selected because it is a well-established algorithm for binary classification. It estimates the probability of an observation belonging to a particular class and provides a relatively straightforward approach for predicting whether a player will have at least one recorded injury in the following season.

4.4 Random Forest

Random Forest was selected because it uses an ensemble of decision trees and can capture more complex relationships between input features and the target variable.

Random Forest also provides feature importance measures, which can help investigate which variables contribute most to the model's predictions. This supports the feature analysis component of the project and provides additional insight into how the model uses the available data.

4.5 Algorithms comparison

The two algorithms were selected because they provide complementary approaches to the same prediction problem. Logistic Regression offers a comparatively simple and interpretable model, while Random Forest offers greater flexibility in modelling complex patterns.

Comparing these models allows PlaySafe to investigate whether the additional complexity of an ensemble approach improves predictive performance on later seasons. It also enables the project to consider different aspects of model quality, including precision, recall and F1-score, rather than relying on accuracy alone.

5. Design Limitations and Future Improvements

Several limitations affect the current design and should be considered before further deployment.

5.1 Data Reliability and Completeness

The dataset was obtained from a non-official source, and its completeness cannot be guaranteed. Missing injury records may represent missing information rather than an absence of injuries.

Consequently, the target variable reflects whether an injury was recorded in the available data, not necessarily whether every real-world injury occurred and was captured.

Future development should investigate the use of more complete and reliable data sources, subject to appropriate access permissions and privacy safeguards.

5.2 Limited Feature Availability

The model uses player characteristics, playing exposure and recorded injury information. It does not include every factor that may be relevant to injury outcomes, such as detailed training workload, medical assessments or other contextual information.

Additional features could potentially improve the model, but they should only be incorporated when their quality, relevance, availability at prediction time and appropriate use have been established.

Also the same players appear in the training and test seasons, so the model could be recognising players instead of learning general pattern.

6. Conclusion

PlaySafe was designed as a supervised machine learning solution for predicting whether a football player will have at least one recorded injury in the following LaLiga season.

The architecture combines historical data, preprocessing, feature engineering, model training, chronological evaluation and risk prediction. The selected features represent player characteristics, playing exposure and recorded injury history. Logistic Regression and Random Forest were evaluated following an initial k-NN modelling iteration.

Overall, the current design provides a foundation for a data-driven decision-support prototype. Its future development should focus on improving data reliability, evaluating

generalisation, validating risk thresholds and confirming that the solution meets the practical requirements of its intended users.